Amplicon sequencing applied in practice

Two applied examples highlighting how amplicon sequencing supports routine marker surveillance and field-based decision-making.

CASE STUDY 1

Drug-resistance and vaccine-target surveillance using nanopore sequencing

Approach

End-to-end targeted nanopore workflow on VWB and DBS — multiplexed amplicon sequencing of *pfcrt*, *pfdhfr*, *pfdhps*, *pfmdr1*, *pfkelch13*, and *pfcsp*, analysed locally with the Nextflow-based nano-rave pipeline.

Relevance to this section

Multiplexed amplicon sequencing used for focused, marker-based questions on field-collected samples, including DBS.

CASE STUDY 4

Therapeutic efficacy of artemether–lumefantrine

Approach

Modified WHO TES protocol in children aged 6–59 months (n=100) in Nchelenge District, a perennial high-transmission setting. Paired baseline and recurrent infections were genotyped with *msp1* and *msp2* plus Illumina amplicon deep sequencing (SeekDeep), with targeted MIPs across 45 drug-resistance loci and *pfmdr1* copy-number estimation.

Relevance to this section

Amplicon deep sequencing of a single highly polymorphic locus used to resolve haplotypes where multiclonal infection is frequent.

Full records for both: Case Studies 1 and 4, Section 13.